Rabiul Awal

Department of Computer Science and Operations Research
University of Montreal, QC, Canada
Mila - Quebec AI Institute, QC, Canada

Email: awalrabiul6@gmail.com
Website: rabiul.me
Research Interests: Generative Models, World Models, RL

Current Position

09/2024 – present **Mila - Quebec AI Institute, Montreal**
Graduate Researcher

05/2026 – present **Qualcomm Research, Amsterdam**
Research Intern

Education

09/2024 – present **University of Montreal**
Ph.D. in Computer Science
Department of Computer Science and Operations Research
Advisor: Aishwarya Agrawal
Courses: Reinforcement Learning and Optimal Control, Vision and Language

01/2021 – 12/2022 **University of Saskatchewan**
MSc. Computer Science
Advisor: Michael Horsch, Roy Ka-Wei Lee

03/2014 – 12/2018 **Noakhali Science and Technology University, Bangladesh**
BSc. Computer Science

Previous Experience

07/2024 – 09/2025 **ServiceNow Research, Montreal**
Visiting Researcher in the Multimodal Team
Advisor: Sai Rajeswar

01/2023 – 08/2024 **Mila - Quebec AI Institute, Montreal**
Research Intern
Advisor: Aishwarya Agrawal

06/2020 – 11/2022 **Singapore University of Technology and Design**
Student Researcher at Social AI Studio
Advisor: Roy Ka-Wei Lee

12/2019 – 12/2020 **Goava Sales Intelligence AB, Dhaka**
Software Engineer

In-progress Research

[4] *UNITE-MoT: Mixture-of-Transformers for Unified Latent Generation* Tackles gradient conflict in joint tokenization–denoising by decoupling FFNs while sharing attention, enabling stable end-to-end training without staged pipelines or external decoders.

[3] *Dynamics Forcing for Diffusion World Models* Addresses weak temporal dynamics in diffusion video models by borrowing self-supervised learning signals to enforce consistent latent transitions.

[2] *Next Embedding Prediction Is World Modeling* Unifies representation learning and world modeling via autoregressive next-embedding prediction, removing EMA, auxiliary losses, and explicit regularization.

[1] *Few-Step Diffusion World Models for Control* Reduces high sampling cost in diffusion control by learning policies from few-step latent denoising with lightweight policy–value heads.

Peer-Reviewed Conference Publications

Google Scholar for full list:
`https://scholar.google.com/citations?user=oOcLfUMAAAAJ&hl=en`
*Denotes equal contribution

[C7] Juan A. Rodriguez, Haotian Zhang, Abhay Puri, Aarash Feizi, Rishav Pramanik, Pascal Wichmann, Arnab Mondal, Mohammad Reza Samsami, **Rabiul Awal** + 7 others. Rendering-Aware Reinforcement Learning for Vector Graphics Generation. *Conference on Neural Information Processing Systems* (*NeurIPS*). 2025.

[C6] Saba Ahmadi*, **Rabiul Awal***, Ankur Sikarwar*, Amirhossein Kazemnejad*, Ge Ya Luo, Juan A Rodriguez, Sai Rajeswar, Siva Reddy, Christopher Pal, Benno Krojer, Aishwarya Agrawal. The Promise of RL for Autoregressive Image Editing. *Conference on Neural Information Processing Systems* (*NeurIPS*). 2025.
Code and models available at `https://github.com/mair-lab/EARL`.

[C5] Anikter Didolkar*, Andrii Zadaianchuk*, **Rabiul Awal***, Maximilian Seitzer, Efstratios Gavves, Aishwarya Agrawal. CTRL-O: Language-Controllable Object-Centric Visual Representation Learning. *IEEE / CVF Computer Vision and Pattern Recognition Conference* (*CVPR*). 2025.
Spotlight presentation at Multimodal Algorithmic Reasoning Workshop, CVPR'25.

[C4] **Rabiul Awal**, Mahsa Massoud, Aarash Feizi, Zichao Li + 9 others. WebMMU: A Benchmark for Multimodal Multilingual Website Understanding and Code Generation. *The Conference on Empirical Methods in Natural Language Processing* (*EMNLP*). 2025.
Spotlight presentation at Multimodal Algorithmic Reasoning Workshop, CVPR.

[C3] **Rabiul Awal***, Saba Ahmadi*, Le Zhang*, and Aishwarya Agrawal. VisMin: Visual Minimal-Change Understanding. *The Conference on Neural Information Processing Systems* (*NeurIPS*). 2024.
Contributed talk at SpLU-RoboNLP Workshop, ACL 2024.

[C2] Shravan Nayak, Kanishk Jain, **Rabiul Awal**, Siva Reddy, Sjoerd van Steenkiste, Lisa Anne Hendricks, Karolina Stanczak, Aishwarya Agrawal. Benchmarking Vision Language Models for Cultural Understanding. *The Conference on Empirical Methods in Natural Language Processing* (*EMNLP*). 2024. **Oral Presentation**.

[*C1*] Le Zhang, **Rabiul Awal**, Aishwarya Agrawal. Contrasting Intra-modal and Ranking Cross-modal Hard Negatives to Enhance Visio-linguistic Compositional Understanding. *IEEE / CVF Computer Vision and Pattern Recognition Conference* (*CVPR*). 2024. **Spotlight presentation** at the Open-Domain Reasoning Multi-Modal Workshop, CVPR'23.

Preprints

[*P1*] Aarash Feizi, Shravan Nayak, Xiangru Jian, Kevin Qinghong Lin, Kaixin Li, **Rabiul Awal** + 11 others. Grounding Computer Use Agents on Human Demonstrations. *arXiv preprint arXiv:2511.07332*. 2025.

Selected Awards, Grant & Fellowship

2026 DIRO Excellence Scholarship for the Winter 2026 session (3900$)

2025 DIRO Excellence Scholarship for the Fall 2025 session 2025-2026 (3000$)

2025 Outstanding Reviewer at CVPR

2024 Selected for CIFAR DLRL Summer School at Toronto

2024 Selected for Upper Bound Talent Bursary at Alberta ($1500)

2022 Selected for Alberta University AI Week Talent Bursary ($1500)

2022 D&I Awardee at ACL Virtual Conference

2021 Graduate Teaching Fellowship at University of Saskatchewan, Canada

2016 1st place at NSTU Intra University Programming Contest, Dhaka, Bangladesh

2015 3rd place at CoU Regional Programming Contest (CoURPC), Comilla, Bangladesh

Service

Broader Community

2024–present **Supervision Request Process Evaluation Committee**
Help professors recruit (pre-evaluation of application) msc and phd students at Mila

2025–present **Whiteboard Session Lead Organizer**
Help organize the RL for Foundation Models reading group for internal lab at Mila

2021–2022 **Computer Science Graduate Council MSc Representative**
Helped in strengthening graduate research activities, hosting social events for grad students at University of Saskatchewan

2023–2024 **Language Grounding Reading Group Organizer**
Help supporting running the reading group on vision language at Mila

2021–2022 **Computer Science Graduate Council MSc Representative**
Helped in strengthening graduate research activities, hosting social events for grad students at University of Saskatchewan

2021–2023 **NLPDhaka Reading Group Founder**
A grass-root initiative for fostering ML/NLP research in Bangladesh
Organized ∼35 crowd-source reading sessions to spread AI research in local community

2017–2018 **NSTU Computing Club President**
Led the computing club growth 10X (from 15 to 150 members), conducted training on competitive programming

Research Community

02/2026 **Workshop Organizer at Mila**
World Modeling Workshop

05/2025 **Workshop Organizer at CVPR25**
Vision Language Models For All: Building Geo-Diverse and Culturally Aware Vision-Language Models

2021–present **Reviewer**
AAAI26, NeurIPS25, CVPR25, ECCV24, CVPR24, ACL23, SIGIR22, AACL22, ICDM21, SIGKDD21

2021 **Volunteer**
ICML21, EACL21, ICLR21, NAACL21

Teaching

09/2025 – 12/2025 **Mila – Quebec AI Institute, Canada**
Teaching Assistant
Computer Science Department
Courses: Representation Learning

01/2021 – 12/2022 **University of Saskatchewan, Canada**
Graduate Teaching Assistant
Computer Science Department
Courses: Data Structure and Algorithms, Python, Introduction to Artificial Intelligence

06/2017 – 06/2018 **Noakhali Science and Technology University, Bangladesh**
Teaching Assistant
CSTE 3110 Microprocessor, Micro-controller and Interfacing Lab

Skills

Python (PyTorch, Tensorflow), Java, C/C++, NoSQL (Elasticsearch), AWS (SageMaker, EC2, Lambda)